

# Greenhouse Climate Controller

Match plants to the right climate zone

**At a glance:** Run the greenhouse controls for a plant that cannot complain, only wilt. Plan about 80 minutes for the core block; 4 teams of 4 students.

## LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: controlled environments, temperature, humidity, light, plant needs, tradeoffs (Understand).
- Design a fair test by controlling one variable while holding others constant (Apply).
- Use their own data to decide whether a redesign improved the result (Analyze).
- Defend a final claim with specific evidence (Evaluate).

## MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- plant profile cards: one set per team (4, plus 1 spare)
- control dial boards: one per team (4, plus 1 spare)
- tour clue sheets: one per team (4, plus 2 spare)
- clipboards: one per team (4)
- dry erase markers: shared, 1 to 2 per team

## PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

- 01 Set up one station per team with the materials above, sorted and ready.
- 02 Stage the scoring sheet and any reference values before testing begins.
- 03 No PPE is needed: the tabletop phase is a dry card-and-board game with no chemicals, soil, or water. For the greenhouse tour, confirm the route and off-limits plants and controls with greenhouse staff, and check the roster for plant or pollen allergies.
- 04 Load the activity slides and ready the projected timer.

## MINUTE-BY-MINUTE FACILITATION

### Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Run the greenhouse controls for a plant that cannot complain, only wilt." Take a quick prediction by show of hands.

## Phase 2 Prediction

0:05 - 0:12

Before any materials move, each team writes one prediction and one reason on the handout. No building yet.

## Phase 3 Constraints

0:12 - 0:18

Set the rules: approved materials only, change one variable at a time, record at least one data point before any redesign, and follow the safety notes.

## Phase 4 Build or solve

0:18 - 0:44

Teams work through the steps on the student handout. Circulate, ask what they are testing, and resist solving it for them.

## Phase 5 Match and set

Teams sort plant cards by need, match each plant to a zone, and set temperature, humidity, and light on the dial board using their tour evidence.

## Phase 6 Resolve tradeoffs

Where two plants conflict in a zone, or raising temperature would dry the air too much, each team picks the setting its evidence best supports and notes the tradeoff it accepted.

## Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with evidence, and reset the station. Wipe the dry-erase boards and return all cards, boards, and clue sheets to the bin so the next team has a complete set; there is no PPE to remove and nothing to wash.

## TROUBLESHOOTING

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**Problem:** A team changed several things at once. **Fix:** Have them reset to one variable before the next reading counts; treat it as a data-quality coaching moment.

**Problem:** A team places a plant with no supporting tour evidence. **Fix:** Ask which clue from the tour backs that zone choice; if they have none, send them back to their clue sheet to find or admit the gap before scoring.

**Problem:** Readings vary between trials. **Fix:** Standardize the procedure: same conditions, same technique, average several trials.

**Problem:** A team finishes early. **Fix:** Push the extension prompt below and ask them to quantify their improvement.

## DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

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Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to controlled environments.

Tier 2 (building but not reasoning):

- "What single change could improve your result without changing anything else?"

Tier 3 (ready to extend):

- "Run a second version that isolates a different variable and compare the two with data."

#### DEBRIEF QUESTIONS (PICK 2 TO 3)

- Which plant was hardest to place, and why?
- What tradeoff did you have to accept?
- How do real growers handle conflicting needs?

#### SCORING RUBRIC (100 POINTS)

| SCORING CRITERION       | POINTS     |
|-------------------------|------------|
| Correct setting choices | 30         |
| Use of tour evidence    | 25         |
| Explanation             | 20         |
| Layout efficiency       | 15         |
| Teamwork                | 10         |
| <b>Total</b>            | <b>100</b> |

#### CLEANUP AND SOURCES

Return reusable items to the bin, dispose of consumables, wipe surfaces, and collect score slips.

**SOURCES** Temple Ambler Greenhouse

# Plant profile cards

Each card says what one plant needs. Match it to the greenhouse zone whose tour readings fit, then defend the placement with a clue from the tour.

## Fern

**TEMP**  cool to warm, 16 to 24 °C (60 to 75 °F)

**HUMIDITY**  high, 50% or more; likes damp air

**LIGHT**  shade or bright indirect; never direct sun

**WANTS** Damp air and shade, like a forest floor.

**AVOID** Direct sun and dry air crisp the fronds.

## Moth orchid

**TEMP**  warm, 24 to 29 °C (75 to 85 °F) by day, above 16 °C (60 °F) at night

**HUMIDITY**  humid, 50 to 80%, with some air movement

**LIGHT**  medium; bright but filtered, no direct beam

**WANTS** Warm, humid air and bright, filtered light.

**AVOID** Full direct sun scorches the leaves (the orchid-loves-sun myth).

## Cactus

**TEMP**  warm days, big day-to-night swing, 24 to 34 °C (75 to 90 °F)

**HUMIDITY**  low; dry air and fast-draining soil

**LIGHT**  very high; full, direct sun

**WANTS** Hot, bright, dry air like a desert shelf.

**AVOID** Damp, shady air rots the roots.

## How to read a card

The thick black bar on each scale is the plant's happy range.

**TEMP**  
10 to 40 °C.

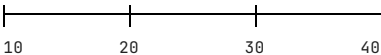
**HUMIDITY**  
0 to 100%.

**LIGHT**  
shade on the left to full sun on the right. A plant in the wrong zone tells you how it suffers: too dry wilts, too bright scorches, too damp molds.

# Control dial board

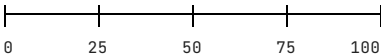
Set each dial for the plant you are placing, then record where each plant goes and the tour clue that backs it.

TEMPERATURE (°C)



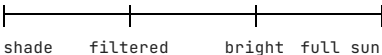
mark your setting with a dry-erase pen

HUMIDITY (% RH)



mark your setting with a dry-erase pen

LIGHT



mark your setting with a dry-erase pen

## Placements: defend each one with a tour clue

| PLANT       | ZONE YOU CHOSE | TEMP | HUMIDITY | LIGHT | TOUR CLUE THAT SUPPORTS IT |
|-------------|----------------|------|----------|-------|----------------------------|
| Fern        |                |      |          |       |                            |
| Moth orchid |                |      |          |       |                            |
| Cactus      |                |      |          |       |                            |
|             |                |      |          |       |                            |
|             |                |      |          |       |                            |

# Tour clue sheet

As you tour, record what you see, feel, and hear for each zone: warm or cool, damp or dry, bright or shaded, plus any reading the docent points out. Back at the table this is your evidence for matching plants to zones.

| GREENHOUSE ZONE | TEMPERATURE | HUMIDITY<br>(DAMP/DRY) | LIGHT<br>(SHADE/BRIGHT) | WHAT THE DOCENT SAID /<br>EVIDENCE |
|-----------------|-------------|------------------------|-------------------------|------------------------------------|
| Main greenhouse |             |                        |                         |                                    |
| Hoop house      |             |                        |                         |                                    |
| Shaded bench    |             |                        |                         |                                    |
| Bright bench    |             |                        |                         |                                    |
|                 |             |                        |                         |                                    |
|                 |             |                        |                         |                                    |
|                 |             |                        |                         |                                    |

Tip: you do not need exact numbers. "Warmer and more humid than the hoop house" is good evidence. Use the control-system or weather-station readout if the docent shows one.